G3 - Prevention, treatment & clinical trials

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Empagliflozin limits AKI incidence and severity following cardiac surgery: an open-label phase IV randomized pilot study

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Background and Aims: Cardiac surgery-associated acute kidney injury (CSA-AKI) is a frequent complication of cardiac surgery assisted by cardiopulmonary bypass (CPB). Currently, no effective preventative strategies exist to reduce the incidence of acute kidney injury (AKI). Sodium-glucose transport protein 2 (SGLT2) inhibitors are effective in reducing the incidence of AKI in studies conducted in patients with chronic kidney disease. Therefore, we hypothesized that perioperative SGLT2 inhibition could reduce the incidence of CSA-AKI.

Method: In this open-label phase IV, randomized, parallel-group, balanced (1:1), pilot study, adult patients undergoing elective cardiac surgery with CPB were randomized to receive 10 mg empagliflozin per os once daily three days before surgery until two days after surgery, or standard care. The primary outcome was AKI incidence according to the Kidney Disease: Improving Global Outcomes (KDIGO) criteria. Other outcomes included hypoxia-inducible factor 1 alpha (HIF1A) and urinary Kidney Injury Molecule (KIM-1), urinary neutrophil gelatinase associated lipocalin (NGAL) and metabolic parameters such as plasma ketone and glucose concentrations.

Results: Between March 2022 and April 2023, 60 patients were randomized to empagliflozin (n = 29) or control (n = 31). All patients who underwent cardiac surgery with CPB were included in the intention-to-treat analysis (n = 25; empagliflozin group, n = 30; control group). SGLT2 inhibition significantly reduced the incidence of all stages of AKI (20% vs 66.7%; absolute difference 46%, 95% CI .13 to .63, $P < .001$). This decline in AKI incidence was reflected in the kidney biomarkers. SGLT2 inhibition also significantly reduced the number of patients with postoperative hyperglycemia (28% vs 60%; absolute difference 32%, 95% CI −56.8 to −7.2, $P = .029$). We observed no differences in the incidence of neither ketoacidosis nor hypoglycemic events.

Conclusion: Perioperative SGLT2 inhibition, compared with standard of care, demonstrated a significant reduction in AKI incidence rates. These findings merit validation in a larger placebo-controlled trial which is currently ongoing.

Supplementary figures:

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Table 1 Kidney biomarkers and incidence of AKI according to KDIGO criteria

	Empagliflozin	Control			
	25	30	Absolute difference	95% CI	P value
Creatinine D1, mean ± SD, µmol/L	98 ± 33	88 ± 34	-9	-28 to 9.1	0.312
Creatinine D2, mean ± SD, µmol/L	97 ± 40	87 ± 39	-10	-31 to 11	0.344
AKI according to KDIGO criteria, n (%)					
Stage 1	3 (12)	11 (37)	-25	.11 to .94	.061
Stage 2	1 (4)	8 (27)	-23	.03 to .82	.031
Stage 3	1 (4)	1 (3)	-1	.13 to 11.1	1.000
All AKI stages	5 (20)	20 (60)	-40	.13 to .63	>.001

Abbreviations: AKI, Acute Kidney Injury; CI, confidence interval; KDIGO, Kidney Disease Improving Global Outcomes; SD, standard deviation

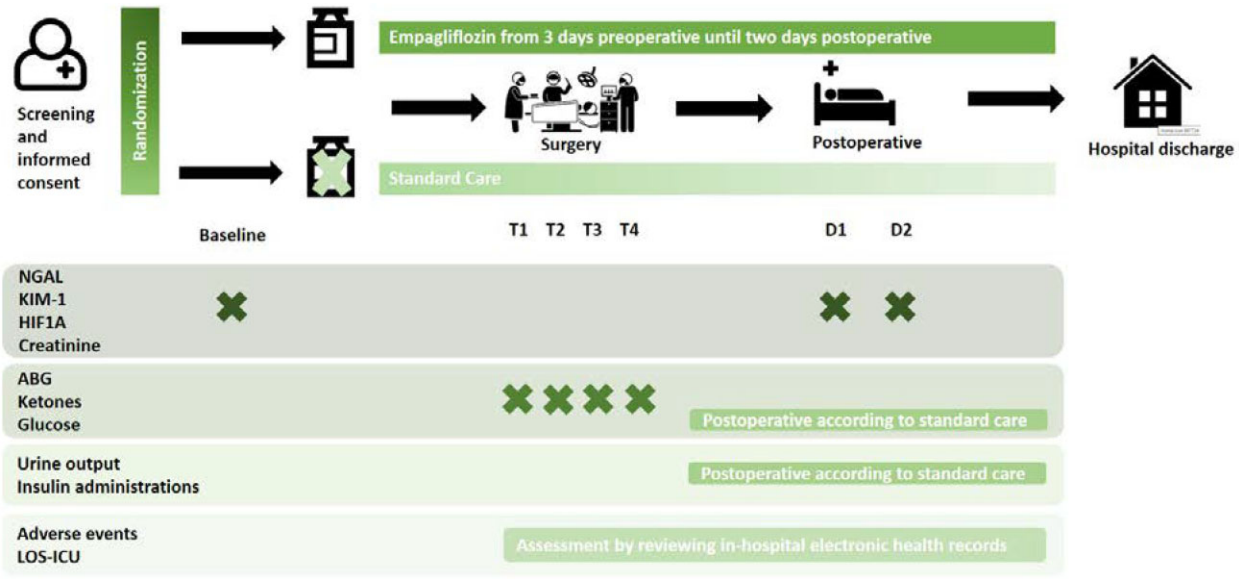


Figure 1: Study flowchart.

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